

Polycentric Privacy Governance in the Digital Age

Zero Knowledge Proofs and the Institutional Privacy Dilemma

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Abstract: Understanding how institutions, technology and social welfare coevolve is essential to understanding economic and social development, and the evolution of privacy alongside institutions and technology in the modern era sheds valuable light on this question. Privacy has both private individual value and public social value, even as privacy rights tradeoff directly with the ability of the government to enforce the law. We deem this inherent defect in the government's incentives to protect privacy the "institutional privacy dilemma," a defect which results in a greater role for technological innovation and individual choice to preserve rights to privacy alongside the public institutions that both protect and infringe it. In the digital age, technological change has led to greater private use of encryption to preserve individuals and organizations' rights to privacy. Yet, this technology itself poses a challenge for law enforcement, for encryption shields those with innocent and criminal motives alike. Unsurprisingly then, governments have sought to weaken or eliminate encryption in the face of the privacy dilemma, with an emerging set of technological solutions preserving individual's rights to a measure of privacy while applying zero-knowledge proofs to simultaneously satisfy legitimate enforcement objectives. Our case study of applications of zero-knowledge proofs and privacy institutions illuminates how a coevolving blend of polycentric forces governs social welfare in practice and emphasizes how individual demand for rights the government is especially likely to infringe plays an essential governance role.

Keywords: privacy, encryption, polycentricity, zero-knowledge proofs, institutional change, technological change, coevolution, complex systems, governance, institutional analysis

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Introduction

Understanding the pivotal role of technological innovation in expanding human potential and societal progress is crucial, yet this relationship is far from straightforward. Instead, it is shaped by a dynamic interplay between technology and institutional frameworks, fostering desired social outcomes while also presenting new regulatory and ethical challenges. Privacy exemplifies this intricate relationship, where three key forces converge in the governance of privacy outcomes: individuals demanding and expecting the respect of their right to privacy, technological advancements enabling and hindering privacy protection, and institutional frameworks safeguarding and compromising privacy rights. This complex interaction continuously shapes and redefines the boundaries of privacy, creating a polycentric blend that is central to modern social and economic development.

Within the polycentric governance of privacy, the “institutional privacy dilemma” refers to the inherent defect in the government’s incentives to protect privacy, given that privacy rights directly trade-off with the ability of the government to enforce the law. Indeed, since the Universal Declaration of Human Rights in 1948, the right to privacy has enjoyed international and constitutional protection in various state jurisdictions worldwide. Yet, the enactment of the Patriot Act marked a significant shift in U.S. surveillance practices, allowing authorities to monitor communications without warrants—a fact brought to public attention by an NSA contractor, Edward Snowden, in 2013. This revelation propelled the adoption of encryption technologies among the general populace after the demand for them became apparent and encrypted applications were provided by software and hardware developers. Despite recently passed legislation enhancing data privacy and continuous advancements in cryptography, government efforts to undermine encryption through demands for backdoors in secure devices persist to date. This conflict underscores a profound tension between governmental surveillance for legitimate security reasons and individuals’ rights to privacy in the digital age.

In response not only to the revelations of this warrantless surveillance of private activity but also to the ever-growing incentives of businesses and corporations to accumulate and profit from user-generated information, came massive growth in the application of cryptography to communications and data (Kuchler, 2024). Once limited to sophisticated computer users, now ordinary individuals can download apps that enable end-to-end encryption of the messages sent back and forth to them. Increasingly, cryptocurrency users can send payments privately in a way that obscures sender and receiver from one another and third parties. Encryption technologies have served to preserve a level of privacy in the digital era that was otherwise diminished given the newfound abilities of the government to warrantlessly monitor citizens’ private correspondence, as well as data collection by private entities in the digital economy.

Yet technological innovation also emerges to resolve dilemmas where sufficient social value is at stake. Recent technological developments involving zero-knowledge proofs (ZKP) offer substantial promise for maintaining privacy while adhering to legal standards. Zero-knowledge proofs have emergent applications in areas as diverse as communication, voting, and financial governance. The characteristics of ZKPs indicate their potential benefits to preserving privacy in the digital age, for these proofs enable the reliable revelation of a fact without revealing the underlying data that validates that fact as true. To give but one motivating example, a financial intermediary can prove its solvency without revealing the specifics of its underlying balance sheets. This example clearly reveals the beneficial tradeoffs embedded in these novel applications of this technology – regulators and consumers can be satisfied with the responsible governance of their intermediary even as that intermediary preserves important private components of its activities.

This paper addresses the polycentric governance of privacy in the digital age. It begins by introducing the polycentric governance of social welfare. Then, it focuses on the polycentric governance of privacy by highlighting three governing forces that co-determine privacy outcomes: private actors, technology, and institutions. Next, it analyzes the institutional privacy dilemma, including the right to privacy in international, regional, and national legal frameworks vis-a-vis existing legislative and regulatory limitations to this right. The analysis proceeds to focus on encryption as an essential element of the institutional privacy dilemma in the digital age, highlighting some recent legislative and enforcement actions that diminish access to encryption, followed by the technical and institutional constraints to successfully banning or weakening encryption. It then presents ZKPs as a potential — partial — solution to the institutional privacy dilemma, highlighting novel applications that could tame the tension between legitimate enforcement motives and disproportionate privacy infringements. Finally, the analysis concludes by considering the generalizable implications of the polycentric governance of social welfare and privacy, specifically highlighting the role of private choice and technological innovation, such as ZKPs, in governing observed privacy outcomes in the digital age.

I. Institutions, Polycentricity and Privacy

Institutions and technology are intertwined and play crucial roles in determining the production possibility frontier and, therefore, the extent of social output that a society can achieve. This section explores how the polycentric governance model applies to social welfare in the specific case of privacy, a social good that individuals highly value and that is shaped by the interplay of individual expectations, technological advancements, and institutional frameworks. We argue that privacy presents unique characteristics compared to other goods and services, as it is affected by a wider range of activities and is more susceptible to disruption by technological innovations. By examining the roles of individuals, technology, and institutions in shaping privacy outcomes, we

introduce the “institutional privacy dilemma” that arises when balancing the protection of individual privacy rights with other competing interests, such as national security and law enforcement.

A. Polycentric governance of social welfare

One implication of institutions mattering for economic development is that they affect the production possibility frontier. Different institutional equilibria can determine the extent of social output that is possible. But technology also quite obviously affects the extent of productive output that is possible within a given society. Institutions and technology are themselves bound together, for one major explanation of the Great Divergence in relative levels of social prosperity surrounds how the institutional context of Europe better protected the rights of inventors (Mokyr, 1992; Koyama and Rubin, 2022). Institutions therefore influence the extent of technological progress that different societies can achieve due to the theoretical and identifiable effects institutions have on incentives to invest scarce effort and capital (Williamson, 2002; Alston and Smith, 2022). Perhaps more obviously, technological change also directly influences institutional development. This makes the pursuit of any private sector output tend to be governed by a polycentric balance of forces in practice; the pursuit of socially valuable objectives (whether revenue-generating or not) is governed by a coevolving blend (Mueller, 2024) of private institutions, public institutions, and technology at any given point in time.

The Bloomington School of institutional analysis is known for, among other foundational concepts, polycentricity (Ostrom, 2010). As a descriptive matter, polycentricity eponymously evokes the reality that in most governance contexts above a certain scale or complexity, outcomes are governed by a blend of nested, overlapping, at times competing, and at times cooperating governance forces. As a normative matter, polycentricity evokes an argument that a balance of polycentric forces is preferable in many governance contexts due to the better governance resulting from such a system. The potential benefits of polycentric governance have been linked to the concept of requisite variety, which holds that a system’s governance should ideally be as complex as the system itself (Lewis and Aligica, 2024). The value of a polycentric lens for understanding privacy outcomes is clear as a descriptive matter; individuals value their privacy as against both other private individuals and organizations, as well as the government itself. As a normative matter, the tension created by the institutional privacy dilemma means that privacy governance will likely be sub-optimal if left largely to public institutional enforcement. This helps shed light on the blend of governance forces that tend to result in privacy outcomes in practice; technological change can both enhance and detract from the level of privacy individuals enjoy, making individual demand for privacy an important source of both privacy-preserving technological change and determinant of privacy outcomes itself, alongside public institutional protection and infringement of privacy.

In sum, outcomes for any social good are co-determined by institutional choice and technological change. The causality likely runs in both directions, just as the effect also runs in both directions. Encryption technology has facilitated privacy even as technological advancements have simultaneously eroded it (Bernal, 2014). Indeed, this example emphasizes how technological progress in one area can beget the need for technological progress in another – the benefits the internet wrought include downsides such as the ability for governments and large commercial entities to collect large amounts of data in ways that weaken traditional boundaries of privacy as against private and public actors alike. Even as technology must keep pace with its own disruptive effects, so too must institutions countenance new technological equilibria. Privacy governance in the digital age is fundamentally polycentric, with private choice, institutional definition, and technological change proving to be fundamental determinants of the delicate balance of privacy outcomes obtained at any given time.

B. Polycentric governance of privacy

Privacy can be understood as a social good that individuals value and institutions and technology shape. Within this polycentric blend of co-evolving governing forces, privacy presents several interesting characteristics relative to other goods and services that people can directly acquire. Privacy displays characteristics of economic rights in that it is tractable to governance by contract and private exchange (Posner, 1977; Acquisti *et al.*, 2015). Yet privacy has also been accorded the status of a human right, and has also been characterized as both a public good (Fairfield and Engel, 2015; Garratt and Van Oordt, 2021), and a commons (Regan, 2002; Mir *et al.*, 2021). This makes its governance especially complex, and, therefore, subject to both the descriptive and normative characterizations of polycentricity (and its basis in requisite variety) defined in the preceding subsection.

First, **individuals** not only value but also expect privacy. Privacy is considered a fundamental civil liberty and a social good (Gavison, 1980). Companies face financial and consumer backlash in the face of large security breaches, and businesses closely guard their information to meet customer expectations and prevent competitors from gaining strategic insights (Acquisti, 2004). This has created a market for company and home security solutions, with the cybersecurity industry growing significantly in the digital age alongside individual identity protection services (Romanosky, 2016). However, despite the reported high value placed on privacy for certain classes of information (Abowd and Schmutte, 2019), individuals appear unwilling to pay for privacy protection in practice to the extent they claim to value it in survey contexts (Acquisti and Loewenstein, 2013; Acquisti *et al.*, 2015). This puzzle can be partly explained by the fact that privacy is enshrined as a right in human rights covenants and modern constitutional orders, leading people to expect it as civil liberty rather than a commodity they must purchase. Just as individuals do not expect to have to pay to

enjoy other negative rights, they may well not expect to have to pay in order to obtain privacy they believe should be accorded as a matter of principle more generally (Kehr *et al.*, 2015; Solove, 2021).

Second, **technological change** has a dual impact on privacy, as it can both enhance and undermine privacy protection. The hyper-connected modern era has brought more granular and comprehensive data about individuals' activities, communications, and whereabouts. One's credit card and bank statements can reveal considerable information about their personal characteristics, health, activities, and preferences. Companies have a strong incentive to collect and retain personal data, as it provides valuable insights into consumer behavior and preferences, enabling targeted advertising and personalized services. Similarly, governments have gained access to unprecedented amounts of personal information, which can be used for surveillance purposes, raising concerns about abuse and the erosion of privacy rights. However, technological advancements like encryption can help protect personal data from warrantless public surveillance and pervasive commercial exploitation. Privacy is affected by a wider range of activities compared to other classes of "siloe" goods and services people value, making it more susceptible to disruption by technological innovations. As technological change can both improve and harm privacy, there is a space for public institutional intervention to preserve privacy.

Finally, **institutions** must balance the protection of individual privacy with other competing interests, such as national security and law enforcement, creating a fundamental tension we term the "institutional privacy dilemma." Human rights covenants and modern constitutional orders enshrine privacy as a fundamental right, like the Fourth Amendment to the U.S. Constitution. However, this right is not absolute and can be subject to certain limitations, particularly concerning public safety and security matters. Governments have often invoked these exceptions to justify surveillance programs, including the NSA's mass data collection revealed by Edward Snowden in 2013. The Supreme Court has been called upon to clarify privacy rights in light of new government surveillance capabilities enabled by technology, as shown in the landmark cases of *Riley v. California* (2014) and *Carpenter v. United States* (2018). This institutional role in defining and protecting privacy rights is further complicated by the rapid pace of technological change, which continually presents new challenges and opportunities for both preserving and infringing upon privacy.

II. The Institutional Privacy Dilemma

In the context of the polycentric governance of privacy, a fundamental tension arises between the public protection of privacy and the public enforcement of the law. The public institutional definition of the right to privacy is widespread in international and national legal frameworks, including human rights covenants, constitutions, and domestic law. Yet the ability to enforce other public institutions requires limitations on the right to privacy for private actors in at least some circumstances. This tension, which we term the "institutional privacy dilemma," is

characterized by the complex interplay of technology, private actor's demand for privacy, and public institutional recognition and limitation of privacy introduced in the first section of the analysis.

A. The right to privacy

From the earliest protections enshrined in global charters, the right to privacy has emerged as a pillar of international, regional, and national legal frameworks. Seen as a cornerstone of human dignity and individual liberty, privacy serves as the bedrock upon which other rights rest, including the freedom of expression, association, and religion. Its scope extends from the corporeal to the informational, safeguarding everything from one's bodily autonomy (protecting against unwanted searches and infringements on decisions about one's health and family) to the inviolability of personal property (shielding homes from warrantless invasions). It also encompasses the confidentiality of our communications — be they via phone, email, or other digital messages — and defends against the unwarranted harvesting and exploitation of personal data.

Globally, Article 12 of the United Nations Declaration of Human Rights enumerates protections against arbitrary invasions of privacy (United Nations, 1948). This mandate is echoed and expanded upon in Article 17 of the International Covenant of Civil and Political Rights, which asserts the entitlement to legal protection against these interferences (United Nations, 1966). Across various regions, provisions like Article 8 of the European Convention on Human Rights (Council of Europe, 1950), Article 11 of the American Convention on Human Rights (Organization of American States, 1969), and Article 17 of the Arab Charter on Human Rights (League of Arab States, 2004) each affirm the right to privacy. Even where not explicitly mentioned, as in the African Charter on Human and Peoples' Rights (Organization of African Unity, 1981), the essence of privacy is implicitly supported through protections against arbitrary acts.

As the digital era continues to unfold, the architecture of international privacy protections has been compelled to evolve and face challenges posed by the expansion of digital surveillance and data collection. The United Nations has proactively curated principles and guidelines (UN Privacy Policy Group, 2018), reinforced by the appointment of a Special Rapporteur on the right to privacy (Office of the High Commissioner for Human Rights, 2015), to steer the global discourse towards enhanced digital privacy norms. Initiatives like the Council of Europe's update to Convention 108 (2018) and the OECD's guidelines (2024) reflect a concerted effort to recalibrate traditional privacy standards to the exigencies of the digital age, aiming to strike a prudent balance between security imperatives and the right to individual privacy.

The right to privacy also appears explicitly or implicitly in nearly every constitution worldwide (Banisar and Davies, 1999). This right is explicitly articulated in the constitution of countries like South Africa, such as in Section 14. However, in other nations, the right to privacy is woven into the fabric of other guaranteed rights. The United States Constitution serves as a prime

example of this. While the term “privacy” is not explicitly mentioned, this right has been reaffirmed through many constitutional amendments. These guarantee the rights to free speech, freedom of religion, and freedom of assembly (First Amendment); protection against the quartering of soldiers in private homes without consent (Third Amendment); protection against self-incrimination and assurance of due process of law (Fifth Amendment); and equal protection under the law, as well as due process, which has been interpreted to include a right to privacy regarding personal decisions and family life (Fourteenth Amendment). Arguably the most robust privacy protection in the U.S. Constitution has come through judicial interpretation of the Fourth Amendment, which explicitly shields individuals from unreasonable searches and seizures by the government. This amendment mandates that law enforcement authorities must, in most cases, obtain a judicially sanctioned warrant based on probable cause before conducting searches in areas where an individual has a reasonable expectation of privacy.

In terms of jurisprudence, the U.S. Supreme Court has played a pivotal role in both affirming and expanding the constitutional foundations of privacy. Landmark cases such as *Riley v. California* (2014) emphasize the modern implications of privacy, ruling that police must obtain a warrant before searching digital information on a cell phone seized at the time of arrest. This decision reflects a broader acknowledgment of the evolving nature of personal privacy in the digital age. Similarly, *Carpenter v. United States* (2018) extended privacy protections further by holding that accessing historical cell site location information also requires a warrant, recognizing the sensitive nature of personal location data and its capacity to reveal intimate details of an individual’s life. These cases illustrate an ongoing adaptation of Fourth Amendment rights, aiming to balance law enforcement objectives with privacy concerns in our rapidly changing technological landscape. Before judicial clarification of the application of Fourth Amendment protections, though, it is notable that the government perspective coming from law enforcement authorities was that searching an arrestee’s cell phone and warrantless collection of cell phone data was constitutionally permissible, notwithstanding Fourth Amendment protections. This tension exemplifies the institutional privacy dilemma that we identify and discuss herein.

Similarly, U.S. jurisprudence has also interpreted privacy, particularly over one’s information or communications, as a right of private actors rather than solely as a right of individuals or natural persons. The landmark Supreme Court decision in *NAACP v. Alabama* (1958) exemplifies this. In this case, Alabama attempted to force the National Association for the Advancement of Colored People (NAACP) to disclose its donor list. While Alabama claimed this was for tax purposes, it was widely understood that the true motives were more insidious, given the state’s history of active Ku Klux Klan presence and pervasive racist attitudes. Without the ability to privately donate, NAACP members would have likely faced significant reprisals, potentially including persecution by the state itself. The Supreme Court ultimately ruled that Alabama’s demand violated the right to due process guaranteed by the Fourteenth Amendment. This case both emphasizes how privacy is a foundation

for other liberties like association, but also underscores how individual rights to privacy comove with the privacy rights accorded to private associations.¹

On the legislative front, the regulatory landscape is dynamically adapting to new and evolving privacy challenges posed by digital technologies. In the European Union (EU), the General Data Protection Regulation (GDPR) is a strong legal framework for data privacy that complements protections in EU member state constitutions (European Union, 2016). The GDPR serves as a robust benchmark for privacy and data protection, offering individuals extensive control over their personal data. This comprehensive framework enhances individual rights and establishes best practices for data protection across various sectors, emphasizing principles such as data minimization, pseudonymization, and the strong encouragement of encryption as a protective measure (Safari, 2017).

Contrasting with the European model, the United States does not have a unified federal privacy law; instead, U.S. privacy governance relies on a more polycentric patchwork of sector-specific laws (Jamison, 2019; Congressional Research Service, 2020). These include the Health Insurance Portability and Accountability Act (HIPAA) of 1996, which safeguards healthcare information, the Children’s Online Privacy Protection Act (COPPA) of 1998 which protects children’s data online, and the Fair Credit Reporting Act (FCRA) of 1970 that regulates the collection of credit information. However, recognizing the gaps in federal legislation, several U.S. states have taken proactive steps by enacting comprehensive data privacy laws. Notably, California has been at the forefront with the California Consumer Privacy Act (CCPA) of 2018 and its more stringent successor, the California Privacy Rights Act (CPRA) of 2020. Virginia’s Consumer Data Protection Act (CDPA) of 2021, the Colorado Privacy Act (CPA) of 2021, the Connecticut Data Protection Act (CTDPA) of 2023, and the Utah Consumer Privacy Act (UCPPA) of 2022 have followed suit (Torres and Zimmer, 2004).

Each of these laws offers similar rights to consumers, including the rights to access, correct, delete, and opt out of the processing of personal data. Importantly, though, most of these state and national privacy institutions define rights against private sector actors, like big technology platforms that tend to collect large amounts of consumer data for advertisement revenue and other purposes. This displays the institutional privacy dilemma that centrally motivates our analysis; while legislatures are responsive to citizen demands for privacy, they appear more responsive when it comes to enforcing privacy rights against other private sector actors compared to self-imposing limits on the government’s own ability to enforce the law or collect tax revenue.

The robust protection of individual privacy rights spans numerous legal jurisdictions, yet has proven a perennial source of debate among prominent legal philosophers and theorists, both historical and modern. One important facet of this discourse concerns the extent and boundaries of privacy rights. Voices of “privacy absolutism” insist on an uncompromised adherence to privacy,

¹ In much of this analysis, the rights to privacy are defined as against the individual, but our arguments apply without loss of generality to the privacy protections accorded to private organizations and individuals alike.

championing the sanctity of personal autonomy. This group includes classic renowned liberal philosophers (Mill, [1859] 1996) and lawyers (Warren and Brandeis, 1890), as well as contemporary thinkers (Solove, 2011; Richards, 2021). In contrast, “privacy pragmatists” propose a more nuanced, context-sensitive approach, willing to adjust privacy expectations and privacy rights under circumstances allegedly considered beneficial for social welfare (Rehnquist, 1974; Goldsmith, 1998; Nissenbaum, 2004). These debates themselves illustrate the inchoate boundary of the “appropriate” level of privacy rights given other public sector objectives. For instance, enforcing revenue collection by tax authorities would be far harder than it is currently if privacy rights lay more toward the absolutist end of the spectrum. The contributions of scholars and thinkers that have significantly shaped the jurisprudential, legislative, and regulatory landscapes over the years are beyond the scope of this analysis, but they emphasize how the governance of privacy in practice is a constantly evolving boundary that responds to citizen demands, public sector objectives, and the status quo of technology which can both afford and degrade privacy.

B. Legislative and Regulatory Constraints on Privacy

Despite the existence of international and national protections for privacy, legislative and regulatory limitations on privacy rights and the usage of privacy-preserving technologies such as encryption have increasingly been proposed and enacted. Legal limitations may define the scope of protected information and set forth the conditions under which privacy can be breached—typically through balancing tests that weigh individual privacy interests against governmental interests like national security or public safety or against other conflicting individual rights (Henkin, 1974, Himma, 2007). Regulatory limitations are imposed through rules and guidelines developed by governmental agencies, which might dictate the specifics of how privacy protections are implemented or bypassed in practice. For example, government surveillance programs, which are authorized under specific legislative acts, are often detailed and expanded through such regulatory guidelines, establishing clear procedures for surveillance activities, including the scope and duration of wiretaps and conditions under which agencies can access private communications.

In practical terms, most government-sponsored mail services are permitted to inspect packages for evidence of criminality. Telephones have also long been subject to wiretaps predicated on a search warrant. However, in the aftermath of 9/11, many Western countries granted law enforcement agencies expanded powers to conduct surveillance, collect data, and monitor communications in the name of counter-terrorism by stating the motive of “preemption of risk” (Webb, 2007). While comparatively slightly less extensive than in the U.S., countries in the European Union implemented surveillance security measures that involved extensive data collection and mining to monitor suspect populations, analyzing behavioral patterns to identify potential threats. These measures aimed to enhance real-time intelligence gathering and preemptively address

security risks, significantly departing from traditional “reactive” policing methods. Examples included the adoption of the European Arrest Warrant, which facilitates quicker extradition processes, allowing for more efficient handling of cross-border terrorism and crime, and the Eurodac, a Europe-wide fingerprint identification system for tracking asylum seekers and illegal immigrants (Levi and Wall, 2004).

With the adoption of the Patriot Act in the U.S., government authorities became able to snoop on citizen’s communications without any warrant, as Edward Snowden’s revelations indicated (Gellman *et al.*, 2013; Greenwald *et al.*, 2013; Karsten and Darrell, 2016). Even more tellingly and consonant with a need for limits on the ability of the government to spy on its own citizens in the first place, information gleaned by national enforcement agencies was then shared with local law enforcement. Because of the secrecy of the means by which U.S. citizens were being surveilled, local law enforcement was also trained to assemble a false investigative history to present to a judge in order to obtain a search warrant for the criminal activity they suspected was occurring, a practice which has been referred to as “parallel reconstruction” (StVincent, 2023).

Although this particular use of private information about citizens and residents of the United States represents an extreme infringement of privacy, it is indicative of the more troubling marginal pattern that we characterize as the institutional privacy dilemma. When it comes to enforcement authorities pursuing their mandates, these authorities’ incentives are not well-aligned to self-policing the boundaries of privacy rights of private citizens: more private information means better law enforcement, *ceteris paribus* (Ashdown, 1981; Rehnquist, 1974). Granted, this type of government action is a major reason for the ability of courts to review government action for consistency with the constitution, but the persistence of surveillance authority — despite the contestable constitutional foundation on which these programs reside (Dowley, 2002; Lilly, 2002; O’Donnell, 2004) — indicates that judicial review of privacy-infringing government activity is likely insufficient to ensure these rights are protected to the extent citizens demand.

The Muslim Pro case is a clear example of public authorities’ demand for private information. Muslim Pro is an application serving more than 100 million users by providing daily prayer times, prayers, and a variety of other content and services tailored to the Islamic faithful. The location data that this application collected was sold through a chain of third parties to the U.S. Army, which admitted to using it in missions of the Special Operations Command (Canaan, 2022). Similarly, ostensibly to enforce immigration policy, U.S. Immigration and Customs Enforcement tracks the movements of drivers in cities home to 75% of the U.S. population, can locate three out of four adults through their utility records, and has access to three out of four of every adult’s driver license data (Wang *et al.*, 2022). The fact that such data is originally collected by state governments underscores the profound challenges faced by authorities when it comes to utilizing data that may exceed the bounds of their authority as compared to 4th Amendment protections. The ease of data collection is, therefore, likely to precipitate further instances where the boundaries of lawful use are

tested, reflecting a fundamental challenge in the self-regulation of government pursuits.² While these aggregated data uses are not subversively motivated, they exemplify the general pattern that given more access to citizens' information (that may be lawfully collected for one government use), government authorities will not hesitate to use that information in pursuit of other purposes. This makes these authorities inadequate to enforce privacy protections: the optimal level of information collection for the government's incentive function is above the level demanded by widespread beliefs in a right to privacy, even if such privacy cannot be absolute. In the digital era, this tension has become particularly acute regarding encryption technologies, a specific example of the institutional privacy dilemma to which the following section is devoted.

III. Encryption as the Digital Age's Institutional Privacy Dilemma

The tension between private actors' rights to privacy and public institutional enforcement motives is itself influenced by technological change. As increasingly granular and aggregated data about private actors' affairs has become available to public authorities, this has reduced the scope of privacy afforded to individuals, which has itself proven to be a source of adoption of encryption applications. This makes encryption technology an illustrative case of how technology, institutions, and private choice coevolve. This dynamic is co-determined by the ability of private actors to encrypt their data, the legislative and regulatory limitations placed on the right to privacy and encryption, recent enforcement actions targeting access to encryption, and the technical and institutional limitations inherent to banning or weakening encryption.

A. Institutional Constraints on Access to Encryption

Legislative adjustments to individuals' privacy rights as before government enforcement motives also involve attempts to ban or weaken encryption. In recent years, U.S. legislative efforts have repeatedly aimed to compel technology companies to provide authorities with the means to decrypt encrypted messages and data (Garfinkel and Spafford, 2022). In 2020, a Republican senator introduced legislation requiring device manufacturers, cloud platforms, and software providers to grant law enforcement agencies access to encrypted data upon presentation of a search warrant. This bill specifically outlined the type of access required for encrypted devices and services, effectively seeking to prohibit end-to-end encryption by mandating the capability to intercept and decode data in real-time (Pfefferkorn, 2020). Although the bill was not enacted, the same senator proposed another bill in 2022 to amend Section 230 of the Communications Decency Act. This amendment would increase the liability of communications providers unless they proactively monitor and filter

² Although the bulk of our illustrative examples of the institutional privacy dilemma are drawn from the United States, the extent of information sharing between the security services of modern liberal democracies, and similar surveillance authorizations in these societies, is evidence that the institutional privacy dilemma is by no means a United States problem.

illegal content on their platforms (Campbell *et al.*, 2022). The problem is that the only way for companies to comply with this would be if they create a means for them to directly observe communications occurring on their platform, which cannot be done without true end-to-end encryption. This concept is known as a “backdoor,” which has long been a point of interest for government authorities concerned about encrypted files and messages.

The extent to which governments have taken a hostile stance toward encryption is not limited to legislative interventions demanding backdoors or criminalizing the use of cryptographic applications without such weaknesses. In 2022, Russian authorities demanded private user information from Telegram. One of the key features of Telegram is its end-to-end encryption option available in “secret chats” which ensures only the sender and receiver can read the messages. However, regular chats use client-server encryption and do not offer the same level of privacy. When Telegram refused to submit private user information, they were fined and access to their service was briefly suspended within Russia (Baghdasaryan, 2022). While this enforcement action is not surprising coming from an autocratic government like Russia’s, authorities in liberal constitutional democracies have also adopted a troubling stance with respect to encryption, well before the Telegram example from Russia. Recent years have provided numerous examples of government attempts to break encryption, many of which have focused on the companies producing such technology themselves (Human Rights Watch, 2018). Famously, the U.S. government demanded a backdoor into Apple’s secured devices following a terrorist attack in California, arguing that such a backdoor was necessary to facilitate a “search” akin to one predicated by a warrant (Electronic Privacy Information Center, 2014).

Enforcement actions have also been brought before the courts in the U.S. and Europe, which again display the negative perspective of public authorities with respect to encryption technology. In 2020, French prosecutors from the National Anti-Terrorism Prosecution Office indicted seven individuals, alleging they were part of a terrorist conspiracy because of their use of encryption technology and other privacy-preserving techniques. The investigation was primarily conducted by the General Directorate for Internal Security (La Quadrature du Net, 2023). The French authorities consistently used as evidence of the alleged criminal conspiracy the fact that the individuals were using encryption tools like Signal to communicate and protect their internet activities by using a VPN and the TOR network. Put more directly, the French government characterized the use of encryption as *de facto* evidence of criminal intent. If widely adopted, such a stance on the part of public authorities will naturally deter citizens from using encryption technologies, even if this makes citizens more vulnerable to criminals themselves, as the subsequent section argues.

Governments’ stances toward the developers of these technologies are even harsher, as recent events surrounding Tornado Cash indicate. This episode began in 2022 when the Dutch authorities arrested Alexey Pertsev, one of the developers of Tornado Cash (Visser, 2022). Pertsev was detained for three months before charges were formally presented and was not granted pretrial release for an

additional five months. Public scrutiny and protests eventually led to his transfer to house arrest. His alleged offense involved creating a protocol that allows for private transactions on the Ethereum network, a protocol that, once deployed, was beyond his direct control. Pertsev’s arrest occurred shortly after the United States imposed sanctions on the network³, and subsequently, other Tornado Cash developers, Roman Storm, and Roman Semenov, faced similar charges in the U.S. (Fannizadeh, 2023). As of this writing, Pertsev was convicted of money laundering in a Netherlands court under the lowest standard of proof in their criminal system, and sentenced to over five years in prison (Greenberg, 2024). While outcomes before the U.S. courts are still pending, these prosecutions are a clear signal to developers of open-source encryption software that they could be liable for any foreseeable illegal uses of their code, even as honest uses for that code also exist to a significant level.

The legal rationale for placing an open-source protocol on a sanctions list is questionable (Reynolds, 2024), and the implications of Pertsev’s public arrest, prolonged detention, and eventual conviction send a clear message to developers of open-source cryptographic software. Specifically, it suggests a potentially expansive liability for third parties’ uses of developers’ technology, irrespective of the geographic location or anonymity of that open-source software’s users. This scenario exemplifies a “chilling effect,” illustrating how such concerns extend beyond merely speech-related issues (Penney, 2022). These actions appear designed to caution other developers and have profound implications for those interested in using privacy-enhancing technologies in the future. As an additional indication of the delicate balance of polycentric forces that govern privacy outcomes in practice, if honest citizens eschew encryption technologies due to fear of criminal suspicion for doing so, this risks becoming a self-fulfilling prophecy as only those with criminal intent become willing to bear the risk of doing so. Problematically, though, there are serious limitations on governments’ ability to prevent citizens from using encryption technologies, which makes the likely incidence of increased costs of access to encryption primarily concentrated on law-abiding citizens. Such enforcement against encryption therefore has a concentrated effect on privacy rights accorded to citizens relative to its intended effect to deter criminal behavior and facilitate law enforcement, a concern the following subsection explores in greater detail.

B. Technical and Institutional Limitations to Banning or Weakening Encryption

Efforts by legislative, regulatory, and enforcement bodies to limit privacy, particularly through restrictions on privacy-preserving technologies, face serious technical limitations. Even when the intention behind these actions may be to enhance national security, public safety, or balance privacy with other individual rights, the practical implications of implementing such measures reveal a number of technical and institutional challenges. Once implemented, the open

³ In August 2022, the Office of Foreign Assets Control (OFAC) at the US Treasury imposed sanctions on Tornado Cash. The sanctions were based on allegations that a group of North Korean hackers utilized Tornado Cash to launder money they had obtained through cyberattacks on U.S.-based cryptocurrency firms and to aid sanctioned government entities in North Korea.

nature of software means encryption tools are likely to remain in the hands of those with criminal intent. The security of backdoors themselves, as well as their responsible custody and use by public authorities, cannot be taken for granted, and in some cases, could operate counter to the public purposes that animate

At a technical level, one of such challenges is the nature of open-source software. The prosecution of the Tornado Cash developers displays an important feature: much cryptographic software is open-source, widely understood, and accessible globally. This ubiquity makes it nearly impossible to restrict access effectively, as demonstrated by the proliferation of “mixers” and “tumblers,” where the application of a privacy-enhancing technology in one context immediately inspires similar applications in others. Even if encryption algorithms were protectable as intellectual property, the very nature of software makes it hard to prevent coders from adapting specific elements of the software to their own purposes. Not only does this make intellectual property enforcement impossible in many cases where software often is not protectable under traditional classes of IP law, but it also indicates that those with criminal intent will continue to be able to develop encryption applications as a function of publicly available information. As large language models reduce the costs of access to information and directly facilitate coding for those who cannot directly write computer programs, the ability of those with criminal intent to access encryption will certainly persist, if not expand.

Another technical limitation is the impossibility of having “secure backdoors.” Much of the current focus of the legislative and enforcement encryption debate is on end-to-end encryption, meaning encrypted data securely transmitted from one user’s device can only be decrypted by the end user’s device. Calls for “secure backdoors” for government access showcase an inherent contradiction that cannot be technically resolved. In providing end-to-end encryption, tech companies cannot ensure private, secure encrypted communications for users and provide law enforcement agencies with a key to decrypt the same conversations. This makes the case of encryption especially useful for understanding the institutional privacy dilemma, for the ability of law enforcement to observe encrypted communications would fully eradicate the benefits of encryption. If the government could ensure that such backdoors would remain exclusively under the control of constitutionally authorized enforcement authorities, this would be limited to being an interesting constitutional question, but the open nature of software suggests that the use of such a tool would be limited to the few authorities that had the narrow constitutional justification to use such an intense intervention into private citizens’ digital lives. Institutionally-mandated backdoors also come with substantial security risks. Such a backdoor is not nearly as simple as providing a lock with a key that only the government possesses and instead presents a clear attack vector for those with malicious intent. As numerous hacks, not to mention ransomware attacks on branches of government, suggest, a government-maintained backdoor could be weaponized frighteningly quickly.

Institutional efforts to ban or weaken encryption also face limitations as to their efficacy. First, such measures would disproportionately impact non-criminals. If criminals can still access open-source cryptographic tools via the open nature of software and the inability of governments to prevent any leaks of their anti-cryptographic tools, this suggests that the primary impact of legal bans and enforcement actions against cryptography will disproportionately affect those without criminal intent. There is clearly a marginal rationale behind the government's approach: making these tools harder to access —though not impossible—might deter some criminals from using them, even if the majority of the burden from such bans falls on the general public. More structurally, the banning or weakening of encryption also directly erodes legitimate justifications for the right to privacy. The often-cited rationale that “if you have nothing to hide, you have nothing to fear” lacks a meaningful limiting factor and can be used to justify any level and scope of government searches of people and property (Solove 2011). Such logic flies in the face of the well-recognized right to privacy that most individuals demand to at least some extent. As individual's private lives in the modern era become increasingly bound up in digital contexts, it is unclear whether a meaningful right to privacy could exist if access to encryption were sufficiently weakened or reduced.

Institutional constraints on encryption could also negatively impact national security. The demand for backdoors in encryption affects civilians and compromises the security of military and governmental communications. Indeed, military and security personnel themselves rely on encryption in communicating, such that a particular Advanced Encryption Standard (AES-256) has come to be called military-grade. However, if the law requires public encryption to include a backdoor, this poses a dilemma: is it possible to ensure that security services themselves can use such encryption without compromise? If tools that can decrypt these communications are developed, the information security services depend on would be at greater risk, especially over foreign networks (Eoyang and Garcia, 2020). There is a further risk of negative international ramifications. Indeed, many other partner nations would likely also request a similar power with respect to encrypted communications within their jurisdictions. The spread of such a general-purpose backdoor would, therefore, be even less likely to remain exclusively in the control of government actors and also raises the more global question of whether all such governments would use such power for legitimate and justifiable reasons.

Access to encryption thus paints the institutional privacy dilemma in stark relief. Private actors' right to privacy afforded to them by encrypted digital activity is in direct, and seemingly dichotomous, tension with the government's intent to enforce a wide range of institutional prescriptions and proscriptions. Absent an additional source of governance, this tension might be altogether irresolvable, forcing one public purpose to more fully give away in furtherance of another. However, technological change as a function of private demands provides another margin by which the social possibility frontier can adjust in the face of private and public demands. As the following

section proceeds to argue, applications of zero-knowledge proofs stand as an emergent example of this phenomenon.

IV. Zero Knowledge Proofs as a Partial Solution to the Digital Institutional Privacy Dilemma

Balancing privacy with other public policy objectives poses an inherent challenge for public institutions. Proposals to ban or weaken encryption also face concrete applicability limitations, either by being technically impossible to achieve or by incidentally leading to the opposite of their intended outcome. Fortunately, ZKPs allow for the revelation of specific data attributes while preventing the wholesale disclosure of the underlying data. ZKPs, while not a complete solution, significantly mitigate the institutional privacy dilemma by reducing the scope of data scrutiny to the minimum necessary while remaining compliant with legal and regulatory requirements (i.e., “functional equivalence”) or by achieving the same outcomes existing laws and regulations aspire to achieve (i.e., “regulatory equivalence”). This section explores the definition of ZKPs and their proliferation with the advent of blockchain technology and smart contracts. It also highlights some current ZKP applications that tame the institutional privacy dilemma in certain enforcement contexts.

ZKPs are a cryptographic method by which a user can demonstrate the validity of a statement to third-party verifiers without revealing information beyond the validity of a given proposition in question itself. In other words, ZKPs enable a user to prove knowledge of a fact without requiring the actual disclosure of the underlying knowledge. For example, in the context of balancing the right to privacy with regulatory requirements such as conducting security checks on public officials, ZKPs can be used to verify that a person meets the necessary criteria for a sensitive job without revealing their personal history. This could involve proving that an individual has no criminal record or the required credentials for a position without disclosing specific details of their personal data, thereby maintaining privacy while still fulfilling security requirements.

The concept of ZKPs first appeared decades ago (Goldwasser *et al.*, 1985), but it was with the advent of blockchain technology and smart contracts that ZKPs applications began to proliferate more extensively.⁴ Specifically, ZKPs have begun to be leveraged on blockchain networks at different levels of the “technological stack”— from the network protocol layer to individual smart

⁴ Blockchain technology enhances the application of ZKPs by providing a decentralized and immutable platform where these proofs can be implemented. The “architectural decentralization” of blockchain (Buterin, 2017) is beneficial for deploying ZKPs with sufficient “confidence” that rules will execute as expected (De Filippi *et al.*, 2020) and privacy of the entities and information being exchanged are crucial. Although outside the scope of our analysis here, this is potentially due to the extent to which blockchain networks provide a credible coordinative institutional backbone for parties that do not necessarily know or trust one another (Alston *et al.*, 2022). On more centralized networks, it is less clear that cryptographically secured data only perceptible through a ZKP-shielding would truly resist tampering if the governing network authorities desired otherwise. The synergy between these two technologies evokes the combinatorial possibilities of the adjacent possible, a concept relevant to understanding social and technological change in complex systems (Deveraux *et al.*, 2024).

contracts and decentralized applications more generally. In doing so, ZKPs show initial promise in anonymizing transactions like payments or votes, helping individuals validate their identity without exposing unnecessary personal information, authenticating certain credentials needed to participate in a specific environment, and reducing bribery or collusion (ethereum.org, 2023). These same applications, which currently are being applied to coordinate private actors in digital contexts, likely can also benefit public compliance.

Certain existing ZKP applications have substantial potential to partly tame the institutional privacy dilemma. On the one hand, these applications can ensure “functional equivalence” by providing a secure and verifiable method for compliance with existing laws and regulations without disclosing irrelevant personal data. For instance, these can allow for verification processes where the specific attributes of individuals are confirmed without exposing the individual’s entire data set. ZKPs applications can also provide “regulatory equivalence” by enabling institutions to achieve the same outcomes existing laws and regulations intend to achieve, even if these applications do so without complying with *specific* legal and regulatory provisions. For example, in financial services, ZKPs can help deter money laundering in a privacy-preserving way by blocking certain transactions without conducting risk assessments as prescribed by existing anti-money laundering provisions (Coalition of Automated Legal Applications, 2021).

The first ZKP application whose compliance benefits we discuss herein is called *Privacy Pools*, which involves the use of ZK-shielded payment networks that can also present proof that the payment did not emanate from certain addresses on the network (Buterin *et al.*, 2023). The job of the United States Department of Treasury’s Office of Foreign Assets and Control, among other regulatory bodies concerned with the application of sanctions, is to identify individuals or groups who should not be allowed to transfer money to one another using the payment networks that comply with U.S. laws and regulations. Provided someone can show their funds did not emanate from, or commingle with, the funds of a blacklisted address, there is no reason to definitively construe such a private transaction as an illegal payment. Given the centrality of sanctions enforcement to government concerns with privacy networks, this alone emphasizes how new technological solutions can help to balance the long-standing tension between legitimate government objectives to peer into the lives of (some) private citizens and these same citizens’ reasonable expectations of privacy in their affairs.

Additional applications of ZKPs also emphasize how they can be applied to facilitate privacy while surfacing information germane to legitimate law enforcement objectives. *zkEmail* enables users to prove that emails have been sent without exposing the content, ensuring confidentiality and integrity in communication (zkEmail, no date). Such a capability could be invaluable in legal contexts, where proving the dispatch of a communication without revealing its sensitive contents could support claims or defenses in civil disputes, fraud investigations, and regulatory compliance scenarios. As pertinently, in the context of national security, zkEmail could be employed to verify

that certain communications between parties include no content that violates security protocols, without revealing the actual content of those emails. This verification process would utilize zero-knowledge proofs to assert the absence (or presence) of these terms without exposing the rest of the email's content, ensuring compliance with security measures while upholding the individual's right to privacy.

In contexts where individuals have voting rights they wish to exercise anonymously and digitally, the governance of these electoral processes faces numerous challenges in securing privacy, validating identity, reducing collusion, and faithfully counting. *Minimal Anti-Collusion Infrastructure (MACI)* solves or strongly mitigates each of these problems, again harnessing ZKPs to beneficial ends, including preserving the privacy of individual voters (MACI, no date). One notable feature of MACI involves the inability of a given voter to prove to anyone else that they voted a certain way, which should reduce the incentives for vote-buying if individuals cannot show to a prospective vote purchaser that they actually voted the way they promised. Given the extent to which secret voting is recognized as a human right worldwide and the persistence of damaging rumors about the integrity of electoral processes (Garnett *et al.*, 2022), this is another way in which ZK-technology ensures privacy while potentially facilitating public sector objectives.

The spectacular implosion of the cryptocurrency exchange FTX in late 2022 emphasizes how important a certain measure of transparency is surrounding the presence of assets custodied by a given exchange (Alston, 2023). Similarly, most stablecoin issuers (i.e., issuers of a type of cryptocurrency designed to maintain a stable value by being pegged to a more stable asset) currently rely on traditional audits of their backing to ensure confidence that their peg is based on convertible liquid assets for the tokens issued (S&P Global, 2023). Indeed, a loss of confidence in Terra also fomented the largest loss in market value in cryptocurrency history, a precipitous decline that itself contributed to the insolvency of FTX (Briola *et al.*, 2023). Once again, a project called *Summa* harnesses ZKPs to be able to reliably signal solvency without revealing individual user balances, number of users, or specific assets and liabilities (Privacy + Scaling Explorations, no date). This proof reveals only that assets exceed liabilities without disclosing other information to which these businesses have well-understood rights to privacy. The applications of this are unlikely to be limited to cryptocurrency exchanges alone, given how vital information about solvency is, even as these businesses have legitimate rights to keep the specifics of their balance sheets private. One additional example beyond exchange functions is that of collateralized stablecoin issuers who may want to reveal balances of certain classes of assets or features of custody environments to users and regulators.⁵

A final concern surrounds surfacing information from a known group while permitting that information to emerge anonymously. Once again, ZKPs have been developed to create a space only accessible to people whose credentials have been validated, but whose posts within that space are

⁵ Tether has consistently argued, consonant with many traditional banking practices, that disclosing the entirety of a bank or financial institution's balance sheet is tantamount to revealing their strategic business decisions to competitors.

fully anonymous. Such is the case of *Semaphore*, a zero-knowledge protocol that enables you to send a message as a verifiable group member while keeping your identity hidden (Semaphore, no date). *Holonym* is another example of a privacy-preserving protocol that enables users to confirm details about themselves, including what their citizenship is (or is not), as well as where their legal residency is (or is not). It does so by issuing non-transferable NFTs, also referred to as “SBTs,” proving specific types of statements (Holonym, no date). The programmability of proofs for negative statements (i.e., “Tom is *not* a U.S. citizen and does *not* reside in the U.S.”) is important. For example, it allows tech-related projects such as unlicensed cryptocurrency exchanges to prove they are not catering to American users without exposing personally identifiable information of the users themselves.

Many of the ZKPs applications mentioned herein are particularly interesting in the context of whistleblowing. Relying on ZKP-anonymized chat applications could provide a credible belief in anonymity on the part of whistleblowers in certain government organizations or high profile private organizational contexts. The more that anonymity can be credibly assured, the more likely it is that whistleblowers will feel confident enough to come forward in the first place. Relatedly, the previously referenced ZK-email technology could also be used to surface emails with specific keywords without the individual’s identity revealing them also being brought to light. This has obvious applications in whistleblower contexts as well.

The range of ZKPs applications reviewed here provides clear examples of compliance with public institutional requirements while facilitating a strong measure of privacy in terms of the underlying data about which only certain characteristics are being revealed. Of course, these applications would not be under development to the extent that they are if it were not for the institutional requirements they are in great part designed to satisfy, nor the underlying demand for privacy that individuals maintain, notwithstanding the large-scale collection of information that the digital age has enabled. ZKPs accordingly provide a useful example of the coevolving polycentric blend of governance forces to which privacy outcomes are subject today.

V. Privacy, Institutional Design, and Technological Change

This narrative should not be misinterpreted as claiming that the government lacks legitimate investigatory motives. Rather, it highlights that regulators rationally seek extensive access to information to maximize the successful pursuit of their lawfully authorized enforcement objectives, and it falls upon concerned citizens and courts to counter this overreach. But if legitimate motives exist, and the alternative is more distortionary and privacy-infringing legal change, this poses a challenge for private citizens who value the affordances cryptography provides them in their correspondence with their loved ones and business associates alike. Our analysis of the polycentric governance of privacy indicates its usefulness as a case by which to better understand the more widespread and perennial coevolution of individual choice, institutions, and technology in

determining outcomes tethered to social welfare in an enduring and complex functional relationship. Technology and institutions matter in determining the extent of social welfare that a given society can obtain through its private and public organizations. Institutions, technology, and social welfare thus coevolve, simultaneously govern, and are governed by one another.

Furthermore, our depiction of the “institutional privacy dilemma” is itself noteworthy on several margins. Better understanding this dilemma partly resolves the puzzle surrounding willingness to pay for privacy in private sector contexts by suggesting reasons that privacy as a fundamental right most subject to infringement by government actors may not be well reflected in the terms of market exchange. It also indicates the fundamentally polycentric governance of privacy in practice. Such governance sits at the dynamic intersection of individual choice that is both constrained and enabled by technological change, which is simultaneously protected and threatened by public institutional protections and enforcement objectives. The specific technical applications of zero-knowledge proofs (“ZKPs”) that we discuss in the preceding section provide a specific lens on these dynamics as they emerge. This privacy dilemma also suggests a critical role for private individuals and technological innovation in preserving civil liberties that the government may be especially likely to infringe upon in pursuit of its public purposes. Consumer demand for and adoption of privacy-preserving technologies alongside private innovation will likely serve as countervailing governance authority centers against the government's justifiable need to observe (to at least some extent) the affairs of the citizens it governs.

Taken as a whole, the analysis herein also highlights the growing entanglement of digital governance questions with more traditional institutional design. Encryption is increasingly inseparable from an individual being afforded a meaningful expectation of privacy in the digital age, emphasizing how material outcomes are increasingly bound up with the digital. This makes a better understanding of both broad polycentric governance forces like technological change and digital institutional precepts in terms of user affordances that correspond to traditional civil liberties, necessary for obtaining good governance in this new digital era. Concepts more frequently cited in computer science and technology journals are now relevant to understanding major contemporary governance questions. Open source software (and its implications), distribution of network control to prevent single points of failure or capture, and access to encryption are all topics with which scholars of institutions and governance should become familiar because institutional design in the modern era has to confront digital governance dilemmas head-on or risk foundering. The contribution herein also identifies micro-institutional foundations of privacy due to the role that encryption will necessarily play in securing this privacy in digital contexts. The more general lesson from this encryption-privacy link is that digital institutional design choices (like the extent of private access to encryption) will increasingly shape outcomes that enhance and constrain our collectively intended private and public purposes.

At a disciplinary level, the field of institutional and organizational economics is founded on the premise that abstracting away the choice of rules in society is insufficient to explain relative variation in performance within and across nations over time (North, 1990). Our necessarily reductive model of the coevolution of institutions, technology, and private choice has a deep implication: taking the technological stock as given for purposes of institutional and economic analysis may ignore the fact that the available technological stock both constrains and responds to the extant institutional and economic environment. Even as technological change both degrades and reinforces individuals' ability to maintain privacy in their affairs, so too does technology respond to the extant consumer and institutional environment that jointly govern privacy outcomes at any given moment. While this dynamic, of course, bedevils clean causal inference of these processes at such a broad level, it also calls into question the viability of assuming technology as given to understand social change, for it is bound up in a complex dynamic with the other forces outlined herein. This is only more the case as the private coordination and public governance of a given society's collective efforts is increasingly entangled with digital associative contexts.

The governance of privacy in the digital age is a valuable case study in the essentially polycentric governance of socially significant outcomes. Liberal constitutional orders have long enshrined privacy as a fundamental right, and the deep value individuals place upon their privacy against private and public sector actors alike is indicated by the extent of modern institutional reification of the right at institutional scales ranging from the private contractual and subnational all the way to the constitutional and international. Yet, the nature of privacy as potentially spanning all an individual's activities, from communication to association to consumption to investment (to name just a few areas!) makes it especially sensitive to disruption by the forces of technological change. It also makes many detrimental changes marginal and, therefore, less cleanly subject to governance by private exchange or public institutional definition. Outcomes in the digital age bring this reality into sharp relief, as governments and network platforms have obtained as much data as they can about private citizens. While this has facilitated law enforcement objectives, this directly exemplifies the institutional privacy dilemma, for the government cannot be relied upon to independently enforce privacy protections in cases where such protections are cut against other public purposes. This has led to citizen demand for encryption of their data and predictable government constraints on access to such encryption. Yet even as public institutions and private choice interact, technology also coevolves with these same forces. New solutions like ZKPs have emerged to preserve rights to privacy while facilitating law enforcement motives.

Individual demand for privacy and the adoption of innovative privacy-preserving technologies is, therefore, likely to continue to play a central role given suboptimal public institutional privacy protection and the detrimental impact of technological change that can facilitate extensive data collection and surveillance. ZKPs and their applications, such as those discussed in this paper and emergent in the future, exhibit considerable potential to alleviate the tension between

legitimate public enforcement motives and excessive privacy losses. By enabling functional and/or regulatory equivalence, ZKPs can facilitate adherence to existing institutions or sufficiently display outcome alignment to institutional intent, while simultaneously minimizing the extent of data scrutiny to what is strictly necessary to prove compliance. Consequently, for privacy governance to sustain outcomes consistent with individual and social valuation of privacy as quintessential civil liberty, private choice, and technological innovation must bridge this persistent gap in order to ensure that privacy protections evolve in tandem with both governmental enforcement and the rapid technological change that carries the near-constant potential to erode individual and organizational privacy.

Conclusion

The foregoing analysis suggests that an understanding of observed privacy outcomes in practice must, at a minimum, acknowledge a trio of forces that codetermine the level of privacy an individual enjoys at any moment in time. Technology will both greatly influence the costs of obtaining information about an individual, but will also provide privacy-enhancing solutions for those who demand them. This makes self-governance also essential, in that individual demand for, and actions undertaken to preserve privacy will determine the uses to which private sector actors can put individuals' data. But the paradox of individuals' being unwilling to pay for the privacy they claim to value suggests privacy may have a deeper social premise than something one must pay for in order to enjoy. This is evidenced by the extent of the public institutional definition of rights to privacy in modern liberal constitutional orders. Privacy rights are, of course, especially salient when it comes to enforcing other public institutions; absent privacy rights, law enforcement would be much easier if authorities could enter households at will and otherwise surveil private actors' activities, including in digital contexts. This essential tension between rights to privacy and public enforcement objectives both greatly defines the balance of privacy protections one enjoys in practice, and means the government will always imperfectly self-police privacy boundaries. As individuals have consistently raised successful claims against state and federal government respondents in major privacy cases, so too will private actors adopt the technical affordances that preserve their privacy. Applications of ZK proofs provide a promising example of these affordances in the digital area, for they facilitate the ability of private actors to prove certain attributes of data or identity without revealing the entirety of the underlying data or identity. Governance of privacy in practice is thus fundamentally polycentric, with private actors, technological capacity, and public government protections and enforcement motives among the most salient spheres of authority determining these governance outcomes in practice.

The profound changes in the U.S.' and other Western States' surveillance practices following 9/11 emphasize the shift towards warrantless monitoring of communications and the consequential

rise in the use of encryption technologies among the general populace. Edward Snowden's revelations in 2013 heightened public awareness and underscored the ongoing tension between governmental surveillance for security purposes and the internationally recognized right to privacy. Despite advancements in data privacy legislation and privacy-enhancing technologies, government demands for backdoors in secure devices continue, constraining the right to privacy and potentially weakening national security. Zero-knowledge proofs can sometimes reconcile the need for government access with preserving privacy rights. The examples mentioned here stand as just a few already developed applications of zero-knowledge proofs, and it seems likely these applications (and some yet to be developed) will play a critical role in balancing the government's legitimate need for information while preserving the privacy rights of individuals in this brave new digital age, marking a potential shift towards a more secure and private framework for digital interactions. In sum, this case study on the emergence of technology to better facilitate individuals' rights to privacy and government law enforcement objectives illustrates the polycentric balance of technology, institutions, and private choice that characterizes more than just privacy rights in the digital age.

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